

FresVeg Direct Farm Harvest Platform

Data Contract & Database Schema Specification

Generated: August 27, 2026 | Version 1.0.0

System Data Architecture & Database Contract

Official Backend Contract Specification for Manager & Database Engineering Team

PURPOSE & SCOPE OF THIS DATA CONTRACT

This document establishes the binding data contract for the FresVeg Agritourism & Direct Farm Harvest platform. It outlines all database schemas, table relationships, field data types, JSONB structures, sample mock payloads, and REST API contract endpoints required for PostgreSQL migration and backend integration.

1. System High-Level Architecture Overview

The FresVeg application operates as a full-stack agritourism and direct-to-consumer farm harvest platform. Key pillars of the data architecture include:

Core Domain Pillar	Key Capabilities & Data Responsibilities
User & Vendor Management	Role-based accounts (Customer, Vendor/Farm Host, Delivery Driver, Admin) with multi-address support.
Agritourism Farm Listings	Rich farm profiles featuring produce, livestock, kids activities, stay accommodations, pricing, and categorized photo galleries.
Agritourism Visit Bookings	Ticket bookings for open tours, payable per-visitor entry, and overnight stay reservations.
Direct Harvest Marketplace	E-commerce catalog for freshly harvested farm produce with delivery or farm pickup fulfillment.
Order & Delivery Fulfillment	Order lifecycle management, driver assignment, live status updates, and tracking numbers.

FresVeg Direct Farm Harvest Platform

Core Database Schemas (Tables 1 - 3)

Generated: August 27, 2026 | Version 1.0.0

2. Primary Database Entities & Data Contract

Entity 2.1: `users` Table (User Accounts & Roles)

Column Name	Data Type	Constraint	Description & Business Rules
id	VARCHAR(255)	PRIMARY KEY	Firestore Auth UID / Unique User Identifier
display_name	VARCHAR(255)	NOT NULL	Full name of user or vendor representative
email	VARCHAR(255)	UNIQUE NOT NULL	Primary email address
phone	VARCHAR(50)	NULLABLE	Contact phone number
role	VARCHAR(50)	NOT NULL	Enum: customer vendor delivery admin
avatar	TEXT	NULLABLE	Profile picture image URL
addresses	JSONB	DEFAULT []	Array of address objects with geocodes
shops	JSONB	DEFAULT []	Array of vendor shop profiles
created_at	TIMESTAMPTZ	DEFAULT NOW()	Timestamp of registration

Entity 2.2: `farms` Table (Agritourism Farm Listings)

Column Name	Data Type	Constraint	Description & Business Rules
id	VARCHAR(255)	PRIMARY KEY	Unique farm ID or slug (e.g. strawberry-paradise)
farm_name	VARCHAR(255)	NOT NULL	Display title of the agritourism farm
location	TEXT	NOT NULL	Address / GPS coordinates for map navigation
description	TEXT	NULLABLE	Detailed farm narrative and guidelines
vendor_id	VARCHAR(255)	FK -> users(id)	Owner user ID of the listing vendor
vendor_name	VARCHAR(255)	NULLABLE	Vendor display name
cost_per_person	NUMERIC(10,2)	DEFAULT 0	Ticket fee per visitor (0 = Free Entry)
cost_type	VARCHAR(50)	DEFAULT free	Enum: free payable
accommodation_price	NUMERIC(10,2)	DEFAULT 0	Base price per night for stay options
accommodations	JSONB	DEFAULT []	Array of stay option objects
crops	JSONB	DEFAULT []	Array of crop strings (e.g. ["Spinach", "Tomatoes"])
fruits	JSONB	DEFAULT []	Array of fruit strings (e.g. ["Mangoes"])
livestock	JSONB	DEFAULT []	Array of animal strings (e.g. ["Gir Cows"])
kids_activities	JSONB	DEFAULT []	Array of kids zone activities
farm_products	JSONB	DEFAULT []	Array of direct farm harvest product objects
crop_photos	JSONB	DEFAULT []	Array of crop field photo objects
livestock_photos	JSONB	DEFAULT []	Array of animal & livestock photo objects
kids_photos	JSONB	DEFAULT []	Array of kids play area photo objects

FresVeg Direct Farm Harvest Platform

Database Architecture & Contract Specification

Generated: August 27, 2026 | Version 1.0.0

Column Name	Data Type	Constraint	Description & Business Rules
accommodation_photos	JSONB	DEFAULT []	Array of stay accommodation photo objects
gallery	JSONB	DEFAULT []	General farm gallery photo objects
visit_days	VARCHAR(255)	DEFAULT Weekends	Operating days (e.g. Mon-Sat)
visit_timings	VARCHAR(255)	DEFAULT 9AM-6PM	Operating hours

FresVeg Direct Farm Harvest Platform

Core Database Schemas (Tables 3 - 5)

Generated: August 27, 2026 | Version 1.0.0

Entity 2.3: `farm\_bookings` Table (Tour & Stay Reservations)

Column Name	Data Type	Constraint	Description & Business Rules
id	VARCHAR(255)	PRIMARY KEY	Unique booking ID (e.g. booking_178523...)
farm_id	VARCHAR(255)	FK -> farms(id)	Target farm listing ID
customer_id	VARCHAR(255)	FK -> users(id)	Booking customer user ID
date	VARCHAR(50)	NOT NULL	Scheduled visit date (YYYY-MM-DD)
visitors_count	INTEGER	NOT NULL DEFAULT 1	Number of guests attending tour
cost_per_person	NUMERIC(10,2)	DEFAULT 0	Ticket cost rate applied
include_stay	BOOLEAN	DEFAULT false	Flag indicating if stay accommodation added
accommodation_title	VARCHAR(255)	NULLABLE	Title of selected accommodation option
rooms_booked	INTEGER	DEFAULT 0	Number of rooms / huts reserved
stay_cost	NUMERIC(10,2)	DEFAULT 0	Total stay fee calculated
total_amount	NUMERIC(10,2)	NOT NULL	Final payable booking amount
status	VARCHAR(50)	DEFAULT confirmed	Enum: pending confirmed completed cancelled
payment_method	VARCHAR(50)	DEFAULT Online UPI	Payment mode used

Entity 2.4: `products` Table (Harvest Marketplace Catalog)

Column Name	Data Type	Constraint	Description & Business Rules
id	VARCHAR(255)	PRIMARY KEY	Unique product ID (e.g. fp-178523...)
name	VARCHAR(255)	NOT NULL	Produce name (e.g. Organic Strawberries)
category	VARCHAR(100)	NOT NULL	Category: Vegetables Fruits Dairy & Eggs Grains
sub_category	VARCHAR(100)	NULLABLE	Subcategory classification
price	NUMERIC(10,2)	NOT NULL	Sale price per unit in INR (₹)
unit	VARCHAR(50)	DEFAULT 1 kg	Measurement unit (kg, pack, crate, dozen)
quantity	VARCHAR(50)	DEFAULT 1	Standard package quantity multiplier
image	TEXT	NOT NULL	Primary high-res product photo URL
farm_id	VARCHAR(255)	FK -> farms(id)	Origin farm listing ID
vendor_id	VARCHAR(255)	FK -> users(id)	Seller vendor user ID
is_deliverable	BOOLEAN	DEFAULT true	False if product is farm pickup only
fulfillment_type	VARCHAR(50)	DEFAULT deliverable	Enum: deliverable non_deliverable pickup_only

FresVeg Direct Farm Harvest Platform

Database Architecture & Contract Specification

Generated: August 27, 2026 | Version 1.0.0

Entity 2.5: `orders` Table (E-commerce Harvest Orders)

Column Name	Data Type	Constraint	Description & Business Rules
id	VARCHAR(255)	PRIMARY KEY	Unique order ID (e.g. order_1785...)
customer_id	VARCHAR(255)	FK -> users(id)	Customer user ID placing order
items	JSONB	NOT NULL	Array of purchased items with price & quantity
total_amount	NUMERIC(10,2)	NOT NULL	Total transaction amount in INR
delivery_address	JSONB	NOT NULL	Full shipping address object
payment_method	VARCHAR(50)	NOT NULL	Payment mode: UPI Card COD
order_status	VARCHAR(50)	DEFAULT placed	Enum: placed confirmed packed in_transit delivered
delivery_boy_id	VARCHAR(255)	NULLABLE	Assigned delivery partner user ID

FresVeg Direct Farm Harvest Platform

JSONB Schemas & Mock Data Contract Payloads

Generated: August 27, 2026 | Version 1.0.0

3. Structured JSONB Data Schemas & Sample Contracts

Below are the exact JSON payload contracts expected by the frontend application for complex JSONB columns.

Sample JSON Payload: Agritourism Farm Payload Contract (farms table)

```
{
  "id": "farm_1785821264",
  "name": "Mannila Organic Paradise & Berry Farm",
  "location": "Mahabaleshwar Highway, Maharashtra 412806",
  "description": "Experience strawberry picking, Gir cow milk tasting, and mud hut stays.",
  "costPerPerson": 250,
  "costType": "payable",
  "crops": ["Strawberries", "Cherry Tomatoes", "Sweet Corn", "Organic Spinach"],
  "fruits": ["Alfonso Mangoes", "Guavas", "Fresh Figs"],
  "livestock": ["Gir Cows", "Free-Range Poultry", "Jamnapari Goats"],
  "kidsActivities": ["Pottery Workshop", "Petting Zoo", "Mini Tractor Rides"],
  "accommodations": [
    {
      "id": "acc-1",
      "title": "Rustic Mud Hut Stay",
      "desc": "Traditional eco mud hut with organic breakfast included.",
      "price": " ₹1,500 / night",
      "icon": "hut",
      "roomQuantity": "2 Huts",
      "roomCapacity": "4 Persons"
    }
  ],
  "farmProducts": [
    {
      "id": "fp-101",
      "name": "Fresh Organic Strawberries",
      "price": 240,
      "unit": "500 g box",
      "image": "https://images.unsplash.com/photo-1464965911861-746a04b4bca6",
      "isDeliverable": true,
      "fulfillmentType": "deliverable"
    }
  ],
  "gallery": [
    { "id": "g-1", "url": "https://images.unsplash.com/photo-1500937386664", "caption": "Strawberry Fields" }
  ]
}
```

FresVeg Direct Farm Harvest Platform

Database Architecture & Contract Specification

Generated: August 27, 2026 | Version 1.0.0

Sample JSON Payload: Farm Visit Booking Payload Contract (farm_bookings table)

```
{
  "id": "booking_1785901234",
  "farmId": "farm_1785821264",
  "farmName": "Mannila Organic Paradise",
  "customerId": "user_cust_9921",
  "customerName": "Sandeep Mannila",
  "customerEmail": "sandeep@example.com",
  "date": "2026-09-10",
  "visitorsCount": 3,
  "costPerPerson": 250,
  "includeStay": true,
  "accommodationTitle": "Rustic Mud Hut Stay",
  "roomsBooked": 1,
  "stayCost": 1500,
  "totalAmount": 2250,
  "status": "confirmed",
  "paymentMethod": "Online UPI"
}
```

FresVeg Direct Farm Harvest Platform

REST API Contract & Database Migration Recommendations

Generated: August 27, 2026 | Version 1.0.0

4. REST API Endpoint Mapping Contract

API Route Endpoint	Functionality & Frontend Usage Contract
GET /api/farms	Fetch all agritourism farm listings with filters (search, crop, location)
GET /api/farms/:id	Fetch full farm details including crops, livestock, stays, products & gallery
POST /api/farms	Create or update a farm listing (Vendor Dashboard & Edit Mode)
POST /api/farms/book	Create a farm tour visit or accommodation booking
GET /api/farms/bookings/all	Fetch vendor farm bookings for dashboard management
GET /api/products	Fetch marketplace harvest products with category & farm filter
POST /api/orders	Place direct harvest order with items payload & shipping address
GET /api/orders/user/:userId	Fetch order history for customer profile

5. Database Indexing & Migration Directives for DB Engineers

Architecture Directive	Technical Implementation Recommendation
JSONB Indexing	Create GIN indexes on `farms.crops`, `farms.fruits`, and `farms.farm_products` for fast JSON searching.
Foreign Keys	Enforce ON DELETE CASCADE on `farms.vendor_id` and `farm_bookings.customer_id` to maintain integrity.
Numeric Precision	Use NUMERIC(10, 2) for all currency fields (`cost_per_person`, `total_amount`, `price`) to prevent rounding errors.
Upsert Strategy	Use `ON CONFLICT (id) DO UPDATE` for `POST /api/farms` to enable seamless draft saving and editing.

